NOVEL
GROUP

SUPPLY CHAIN MANAGEMENT & QUALITY CONTROL PLATFORM

Functional & Technical Specification

Version 2.0 — BOS-Native Architecture

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Change from v1.0

Odoo is out. Everything is built natively inside Novel BOS. BOS is now the system of record — it owns masters, stock quantity, documents, quality states and postings. Daily invoice upload becomes a primary data-capture channel.

This is a bigger build than v1.0. It is achievable, but only if we sequence it correctly and are honest about what daily invoice upload can and cannot do (Section 3.4). Read Sections 1.4 and 3.4 before estimating.

0. WHY WE ARE BUILDING THIS

Novel is scaling fast across 4 brands, 2 plants and 8+ sales channels. At this scale the failure modes are no longer "we forgot to order" — they are:

1. **Stock-outs on hero SKUs** while dead stock sits at another node.
2. **Quick-commerce fill-rate misses** → penalties, delisting, ranking loss.
3. **No batch traceability.** A single consumer complaint on a baby SKU with no forward/backward trace is an existential risk. Baby care and cosmetics are regulated, litigated categories.
4. **Purchase decisions on WhatsApp and gut feel**, not on a net-requirement calculation.
5. **QC as a signature, not a system** — no sampling discipline, no NCR trail, no vendor accountability.

0.1 Success metrics (the system is only "done" when these move)

Metric	Baseline (capture in Phase 0)	Target — 12 months
Channel fill rate (OTIF to marketplace/QC POs)	TBD	≥ 97%
FG stock-out days on A-class SKUs	TBD	< 1% of SKU-days
Inventory days of cover (blended)	TBD	45–55 days, no SKU > 120
Dead + near-expiry stock as % of value	TBD	< 2%
Forecast accuracy (WMAPE, A-class, monthly)	TBD	≤ 20%
RM/PM purchase price variance vs rate contract	TBD	≤ 1.5%
Incoming rejection PPM (vendor)	TBD	< 5,000, trending down
Consumer complaints per million units	TBD	< 50
Batch trace time, both directions	Days	**< 10 minutes**
POs raised without a system-generated requirement	~100%	< 5%
Stock-ledger accuracy (physical vs system, A-class)	TBD	≥ 99%

1. SCOPE & ARCHITECTURE

1.1 In scope

Plan → Source → Make → Store → Quality → Deliver, plus the transactional and document layer underneath it.

- **Entities:** Novel Tissues Pvt Ltd — build multi-company / multi-GSTIN from day 1

- **Plants:** Mysuru, Manesar — design for N plants
- **Brands:** Novel Babio, ALMA, Novel Cosmetics, Medibath
- **Item classes:** FG, SFG, RM, PM, consumables, spares, combo/kits
- **Channels:** Amazon (FBA / Easy Ship / Seller Flex), Flipkart, Meesho, Blinkit, Zepto, Swiggy Instamart, BigBasket, Tata 1mg, FirstCry, Shopify D2C, GT distributors (UP onward), B2B / private label

1.2 Out of scope (v2 build)

Payroll and HR, statutory return *filing*, full statutory books of account (trial balance, P&L, balance sheet). BOS generates the data and exports to the accounting package (Tally or equivalent); the auditor's books stay there. **Do not build a general ledger.** Everything else — inventory valuation, landed cost, PO-GRN-invoice matching, GST-ready documents — is in BOS.

1.3 BOS module map (all native)

#	Module	Owns
M1	**Master Data**	Items, BOM, vendors, customers/channels, warehouses/bins, tax, UOM
M2	**Transactional Core**	Document engine, numbering, stock ledger, postings, GST docs, e-invoice, e-way bill
M3	**Document & Invoice Ingestion**	Daily upload, OCR/parse, matching, exception queue, GST recon
M4	**Demand Planning**	Sales ingestion, forecasting, consensus, accuracy
M5	**Inventory Policy Engine**	ABC-XYZ-FSN, safety stock, ROP, EOQ, min-max, node policy
M6	**MRP / Auto-PO Engine**	Net requirement, BOM explosion, pegging, PO suggestions
M7	**Procurement & Vendor**	RFQ, rate contracts, PO lifecycle, vendor portal, scorecard, 3-way match
M8	**Inbound & Warehouse**	Gate, GRN, putaway, bins, FEFO, cycle count, STO
M9	**Quality (QC/QA)**	Plans, sampling, IQC/IPQC/FQC, NCR/CAPA, batch release, traceability, recall
M10	**Manufacturing**	MPS, MO, job cards, BMR/BPR, yield, OEE, maintenance
M11	**Outbound & Channels**	Channel PO ingestion, ATP, allocation, pick-pack-dispatch, ASN, fill rate
M12	**Costing**	Landed cost, valuation, variances, SKU × channel margin
M13	**Control Tower**	Dashboards, alerts, reports, exception engine
M14	**Platform**	Auth, roles, audit trail, workflow/approval engine, notifications, integrations, mobile

Foundational rule (was the Odoo rule, now internal): stock quantity exists in exactly one table — the BOS stock ledger. Every other number in the system is derived from it. No module keeps its own quantity. No screen writes a quantity directly; quantity only changes through a posted document.

1.4 What we now have to build ourselves — read this before estimating

Dropping the packaged ERP means BOS must build these, and they are unglamorous but unavoidable:

6. **Document engine & numbering** — series per document type per company per FY, gapless, non-editable, with cancellation (never deletion).
7. **Double-entry stock ledger** — every movement is a paired debit/credit against locations. Balances are always derived by summing the ledger, never stored as a mutable field. This is the single most important design decision in the build; get it wrong and the system is unrecoverable at year two.
8. **Period close & locking** — once a month is closed, no back-dated postings. Corrections via reversal documents.

9. **Inventory valuation** — moving weighted average per item per node, with a valuation ledger parallel to the quantity ledger.
10. **GST-compliant documents** — tax invoice, credit/debit note, delivery challan, job-work challan, with correct place-of-supply, HSN, reverse charge handling.
11. **E-invoice (IRN) and e-way bill** via a GSP. Do not build against the government portal directly.
12. **Approval/workflow engine** — configurable, reusable across POs, NCRs, adjustments, batch release.
13. **Audit trail framework** — global, append-only, applied to every table by default rather than per-module.
14. **Accounting export** — a clean, reconciled hand-off to Tally: purchases, sales, debit/credit notes, stock valuation, expenses.

Estimate impact: this is roughly 10–14 additional developer-weeks before a single SCM feature is visible. It is not optional and it cannot be retrofitted. Budget for it in Phase 0/1 or the project will stall at month four.

1.5 Tech stack (Sumukha to confirm)

- Backend: Python (FastAPI) or Node (NestJS) — one language, one framework, no mixing
- DB: PostgreSQL 15+ (partition the stock ledger and audit log by month from day 1)
- Jobs: Celery/BullMQ + Redis for MRP runs, forecast runs, ingestion, alerts
- Frontend: React + Tailwind; PWA for all mobile/scan flows
- Forecasting: Python, statsmodels / scikit-learn — explainable models only, no black box in v1
- OCR/parsing: layout-aware extraction for invoice ingestion (Section 3)
- Integrations: Amazon SP-API, Flipkart, Meesho, QC platform portals, Shopify Admin API, GSP for e-invoice/e-way bill, WhatsApp Business API
- **Hosting: the current Hostinger box is not adequate.** Move to managed cloud (AWS ap-south-1) with daily automated backup, point-in-time recovery, and a tested restore before go-live.

2. MASTER DATA — M1

> 80% of SCM system failures are master data failures. Phase 0 is master data. Nothing else starts until SCM Head and QA Head sign it off.

2.1 Item master

- Item code (structured: `BRAND-CATEGORY-VARIANT-PACK`), legacy codes, EAN/GTIN, channel IDs (ASIN, FSN, Meesho, platform SKU codes)
- Class: FG / SFG / RM / PM / Consumable / Spare / Kit
- Brand, category, sub-category, variant, pack size; pack config (units → inner → case → pallet)
- Base UOM + alternate UOMs + exact conversion factors
- Net weight, gross weight, L×B×H, volumetric weight, pallet pattern
- HSN, GST rate, MRP with **effective-dated MRP history** (Legal Metrology requirement)
- Shelf life (months) and **minimum residual shelf life per channel** (quick-commerce platforms reject short-dated stock — hard field, e.g. ≥ 70% life remaining)
- Storage conditions, hazard class
- **QC linkage:** inspection plan ID, sampling plan, COA required Y/N, retention sample Y/N
- Sourcing: make/buy, primary vendor, Approved Vendor List — **only AVL vendors can be POed**

- Planning: ABC / XYZ / FSN, lead time, MOQ, order multiple, shelf-life-capped max order qty
- Status: NPD / Active / Phase-out / Discontinued / Blocked (with reason); launch and planned EOL dates

Item creation workflow: requester → Category/NPD → Costing → **QA (inspection plan must exist)** → Finance (HSN/GST) → Active. No item goes live without a QC plan. Non-negotiable.

2.2 BOM & routing

- Multi-level (FG ← SFG ← RM/PM), effective-dated versions, no in-place edits
- Line fields: component, qty per, UOM, scrap/wastage %, phantom flag, approved alternate with substitution rule and QA approval
- Routing: operations, work centre, machine, cycle time, setup time, manpower standard
- Change control via ECN with QA sign-off, effective date, and disposition of existing old-component stock
- Costed BOM roll-up: standard cost vs latest purchase cost, with variance view

2.3 Vendor master

- Code, legal name, GSTIN, PAN, **MSME/Udyam status** (drives the 45-day payment rule — flag it), addresses, bank details
- **Bank detail changes require dual approval plus a call-back verification record.** This is the single biggest fraud vector in procurement; build the control now.
- Vendor-item mapping: price, lead time, MOQ, pack, rate contract validity, price-break slabs, escalation clause
- Payment terms, credit days, incoterms
- Qualification: audit date and score, GMP/ISO certificates with expiry — **expired certificate auto-blocks the vendor from new POs**, alert at 60 days
- Status: Prospect / Under approval / Approved / Conditional / Blocked / Blacklisted
- Scorecard rollup (Section 8.4)

2.4 Warehouse / location / bin

Hierarchy: Company → Site (Mysuru, Manesar, external FCs, distributor DCs) → Zone → Rack → Bin.

- Every bin barcoded, with volume and weight capacity, storage type, temperature class
- **Mandatory zones, physically and systemically separate:** `QUARANTINE`, `REJECTED`, `RETURN-TO-VENDOR`, `NEAR-EXPIRY`, `MRB HOLD`, `RETENTION SAMPLES`
- Virtual locations for: Amazon FBA, Flipkart FC, Meesho, quick-commerce dark stores, distributor stock, goods-in-transit, job-work-at-vendor

2.5 Channel / customer master

Per channel: PO ingestion method, lead time, appointment rules, label and barcode spec, packaging spec, minimum residual shelf life, penalty structure, returns policy, payment terms, SLA.

2.6 Master data governance

Named data owner per master. Every master change is workflow-approved and audit-logged. Bulk upload templates with server-side validation and a rejection report — never a silent partial import.

3. DOCUMENT & INVOICE INGESTION — M3

This is the module that makes "upload invoices daily and BOS captures everything" work. It has to be built properly or it becomes a data-quality sinkhole.

3.1 What gets uploaded, and how

Stream	Source	Frequency	Method
Purchase invoices (RM/PM/services)	Vendors	Daily	Vendor portal upload (preferred), email-to-BOS inbox, bulk drag-drop, WhatsApp-to-BOS
Sales invoices (channel + GT + B2B)	Own dispatch team	Daily	Generated *in* BOS (preferred) or uploaded
Channel settlement / payment reports	Amazon, Flipkart, Meesho, QC platforms	Daily/weekly	API or file upload
Transporter LR / freight bills	Transporters	Daily	Upload
Vendor COA / test certificates	Vendors	Per lot	Vendor portal, linked to the batch
GSTR-2B	GST portal	Monthly	File upload for ITC reconciliation
Distributor secondary sales & stock	GT distributors	Weekly	Template upload or distributor app

3.2 Ingestion pipeline

```
Upload (portal / email / API / bulk)
→ Store original file immutably (S3-equivalent, hash-stamped, never overwritten)
→ Classify document type
→ Extract (OCR + layout parsing; native parse for PDFs with text layer; direct parse for XML/JSON e-invoice)
→ Normalise (vendor GSTIN → vendor code; item description + HSN → item code via a learned mapping table)
→ Validate (GSTIN format, tax arithmetic, HSN vs item master, duplicate invoice check)
→ MATCH against the open PO / GRN / dispatch
→ Auto-post if clean, else route to the EXCEPTION QUEUE with the reason
→ Human resolves exception; the resolution trains the mapping table
```

3.3 Matching rules

- **Purchase invoice:** match on vendor + PO number + item + qty + rate. 3-way match against PO and GRN accepted quantity (post-QC). Tolerances default to 0% on both qty and rate; anything outside blocks and queues.
- **Duplicate detection:** vendor GSTIN + invoice number + FY is a hard unique key. Reject duplicates loudly.
- **Unmatched invoice:** parked in the exception queue with an ageing clock. Weekly report of invoices unmatched > 7 days to SCM Head; > 15 days to AP.
- **Item mapping table:** vendor's description ↔ our item code, learned from confirmed matches, editable, with confidence score. Below a confidence threshold it always goes to a human.

3.4 What invoice upload CAN and CANNOT do — please read

It can: capture spend, prices, tax, vendor performance on paper, sales value, channel settlements, ITC reconciliation, and give you a complete financial picture with one day's lag.

It cannot:

- Hold **quantity** accurately. An invoice says what was billed, not what physically arrived, passed QC, or was put away. Quantity must come from a scanned GRN.

- Give you **batch, MFG date, expiry**. These come from the physical receipt, not the invoice. Without them there is no traceability, no FEFO, no recall, no QC module — the entire Section 10 collapses.
- Give you **quality state**. Quarantine vs released is a decision, not a document.
- Support **MRP**. Auto-PO needs real-time on-hand, in-transit and reserved quantities. A one-day-lagged invoice feed cannot drive a reorder point.
- Capture **production, consumption, wastage, yield** — no invoice exists for these.

Therefore the operating model is:

> **Physical events are transacted live in BOS on the scanner** (GRN, putaway, QC, issue, production, pick, dispatch, count). **Documents are uploaded daily** to enrich, value and reconcile those events.

Invoice upload is the *financial and reconciliation* spine. Scanning is the *physical* spine. The system needs both; either alone gives you a half-system. This is the one point in the whole design I'd push back on hardest — if the plan is invoice-upload-only, drop the QC module and MRP from scope, because they cannot work.

3.5 Exception queue

A single work-list screen: document, type, source, reason for exception, ageing, assigned owner, action. Bulk actions supported. **The health of this queue is the health of the data.** Put its size on AP's daily dashboard.

---

4. TRANSACTIONAL CORE — M2

The layer the packaged ERP would have given us. Build it first; everything sits on it.

4.1 Document engine

- Every business event is a document with: type, series, number, date, company, status (Draft / Submitted / Cancelled / Amended), created-by, approved-by, references, attachments
- Gapless numbering per type/company/FY, server-generated, never editable
- Draft → Submitted (posts to ledger) → Cancelled (reversing entry) or Amended (new version, old retained). **No document is ever deleted or silently edited.**
- Document types: PO, PO amendment, GRN, purchase invoice, debit note, stock transfer, delivery challan, job-work challan, manufacturing order, material issue, production receipt, stock adjustment, scrap note, sales order, dispatch, tax invoice, credit note, return receipt, QC inspection lot, NCR, batch release

4.2 Stock ledger (the core table)

```
stock_ledger_entry:
  entry_id, posting_datetime, document_type, document_id, document_line
  item_id, batch_id, from_location_id, to_location_id
  quantity, uom, quality_state, rate, value
  created_by, created_at (server time, immutable)
```

- Every movement is a paired location debit/credit — receipts come *from* a virtual `SUPPLIER` location, dispatches go *to* a virtual `CUSTOMER` location, so the ledger always balances to zero. Losses go to a `SCRAP`/`VARIANCE` location, never vanish.

- **Balances are always computed by summing the ledger.** Maintain a materialised balance table for performance, rebuilt nightly and reconciled — but the ledger is truth. If they ever disagree, the ledger wins and an alert fires.
- Append-only. No updates, no deletes. Corrections are reversing entries.
- Partition by month; index on (item, location, batch), (posting_datetime), (document).

4.3 Valuation ledger

Parallel to the quantity ledger. Moving weighted average per item per node. Landed cost apportionment (Section 14) posts here. Produces closing inventory value by item, batch, node, quality state, and ageing bucket.

4.4 Period control

Monthly close: lock postings before the close date; back-dated entries rejected. Reopening a period requires AP approval and is logged.

4.5 Statutory documents

- GST tax invoice, credit and debit notes, delivery challan, job-work challan (with the 143 one-year return clock tracked)
- Correct place-of-supply logic, CGST/SGST/IGST determination, reverse charge, TCS on marketplace sales
- **E-invoice IRN generation via GSP** with QR, retry on failure, cancellation window handling
- **E-way bill generation and Part-B update**, extension and cancellation
- HSN-wise outward and inward summaries; GSTR-1 and GSTR-3B data extracts; GSTR-2B vs purchase register reconciliation with a mismatch worklist

4.6 Accounting export

Reconciled daily/monthly export to Tally: purchases, sales, credit/debit notes, expenses, stock valuation, with a control total on each batch and a re-push facility. **Never a one-way fire-and-forget export — always with acknowledgement and a reconciliation report.**

4.7 Workflow & approval engine

Configurable, reusable: entity + condition (value, item class, vendor status, deviation %) → approver role(s) → sequential or parallel → escalation on SLA breach → full trail. Used by POs, adjustments, batch release, NCR disposition, master changes, MRP overrides. Build once, use everywhere.

4.8 Audit trail framework

Applied by default at the ORM/data-access layer to every table: who, what, when, old value, new value, reason. Append-only, queryable, exportable for audits. Retention: shelf life + 5 years for anything quality-related.

5. DEMAND PLANNING & FORECASTING — M4

5.1 Data ingestion (daily, automated)

- Primary sales: channel × SKU × day × FC (Amazon by fulfilment centre — critical for placement)
- Secondary sales: distributor sell-out, quick-commerce dark-store sell-out
- Stock at every node: own WH, FBA, marketplace FCs, dark stores, distributor, in-transit

- **Lost-sales proxy:** out-of-stock days, buy-box loss, listing suppression. *Do not forecast on shipped quantity alone — you will systematically under-forecast exactly the SKUs that matter most.*
- Price and promo calendar, ad spend by SKU
- Market signals from the existing rank-tracking automation

5.2 Forecast engine

- Cleanse: outlier detection, OOS-corrected demand (impute demand during stock-out periods)
- Models per SKU-channel, auto-selected on backtest error:
  - Moving / weighted moving average — short history, new launches
  - Holt-Winters exponential smoothing — seasonal SKUs
  - Croston / SBA — intermittent demand (GT, B2B, slow movers)
  - Regression with promo, price and ad-spend regressors — A-class SKUs
- **Event overlays:** Dasara/festive, Big Billion Days, Great Indian Festival, Prime Day, Republic Day, monsoon seasonality (wipes/diapers), school reopening. Maintain an event calendar master with historical uplift factors per SKU class.
- New product forecasting: analogue-SKU method (reference SKU + ramp curve), replaced by actuals after 8 weeks
- Horizon: 13 weeks daily-bucketed + 12 months monthly, refreshed weekly (Monday 06:00 IST)

5.3 Consensus process (system-enforced, weekly)

15. **Tue** — statistical forecast published to Sales/KAMs
16. **Wed** — KAM override with mandatory reason code (promo / listing change / competitor OOS / new geo / customer commitment) and % delta captured
17. **Thu** — supply review; constrained plan generated
18. **Fri** — consensus locked; AP signs off A-class deviations > 20%
19. Freeze rules: weeks 0–2 frozen, 3–6 slushy, 7+ free

5.4 Accuracy tracking

WMAPE, MAPE and BIAS per SKU-channel-planner-KAM. **Monthly scorecard published by name.** Forecast accuracy without accountability is theatre.

6. INVENTORY POLICY ENGINE — M5 ("minimum stock holding")

Policies are **computed, not typed**. Recalculated monthly or on demand; auto-applied for B/C class, approval-gated for A class.

6.1 Segmentation

- **ABC** by annual consumption value (A ≈ top 70–80% of value, B next 15%, C rest)
- **XYZ** by demand variability, coefficient of variation (X < 0.5 stable, Y 0.5–1.0, Z > 1.0 erratic)
- **FSN** by movement frequency

	X (stable)	Y (variable)	Z (erratic)
A	SL 98%, continuous review, min-max	SL 98%, weekly review, higher SS	SL 95%, make-to-order, tight watch
B	SL 95%, min-max	SL 95%, periodic review	SL 90%, order on demand
C	SL 90%, bulk buy, long cover	SL 90%	SL 85%, no stock / on demand

6.2 Formulas — implement exactly

Safety stock (demand + lead-time variability):

```
SS = Z × √( LT × σd²  +  d̄² × σLT² )
```

- `Z` from target service level: 95% → 1.65 | 97% → 1.88 | 98% → 2.05 | 99% → 2.33
- `d̄`, `σd` = mean and std dev of daily demand
- `LT`, `σLT` = mean and std dev of lead time — **computed from actual GRN date minus PO date. Never use the vendor's claimed lead time.**

Reorder point: `ROP = (d̄ × LT) + SS`

EOQ: `EOQ = √( 2 × D × S / H )` — D = annual demand, S = ordering cost per order, H = annual holding cost per unit (use 18–22% of unit cost for India: cost of capital + storage + obsolescence)

Max stock: `Max = ROP + EOQ`, hard-capped by shelf life:

```
Max_allowed = d̄ × (Shelf_life_days × 0.6)      // never hold more than 60% of shelf life
```

Days of cover: `DOC = Closing stock ÷ trailing-28-day average daily run rate`

Order-up-to level (periodic review): `S = d̄ × (LT + Review_period) + SS`

6.3 Node-level policy — where most companies fail

Policy is set **per SKU per node**, not per SKU:

- Amazon FBA — cover target by FC, respecting IPI and restock limits
- Flipkart FC, Meesho
- Quick-commerce dark stores — 3–7 days cover, replenished 2–3× weekly, very tight
- Plant / CFA — covers the gap and buffers channel volatility

Network cover is the sum, and it must be visible on one screen. Otherwise you buy stock that already exists 400 km away.

6.4 Redistribution engine

Daily job: find (SKU, node) pairs where node A is above max while node B is below ROP → suggest an STO with cost/benefit (transfer cost vs stock-out cost) → approval → STO document with e-way bill.

6.5 Alerts

Below ROP | below SS | projected stock-out inside the lead-time window | above max or cover > 120 days | zero stock on an active A-class SKU (escalate to AP) | near expiry at 6/4/3/2 months | no movement 90/180 days.

7. MRP / AUTO-PO ENGINE — M6

The core ask. Runs nightly at 02:00 IST and on demand.

7.1 Net requirement (time-phased, weekly buckets, 26-week horizon)

For every item × plant × bucket:

```
Gross Requirement = Independent demand (FG forecast + firm channel POs + B2B orders)
                  + Dependent demand (BOM explosion from parent planned orders)

Available         = On-hand UNRESTRICTED only
                  − Reserved / allocated
                  − Quarantine / blocked / rejected        // NEVER counted as available
                  + Scheduled receipts (open PO, in-transit, MO output) landing in bucket
                  + Planned STO inbound

Net Requirement   = Gross Requirement − Available + Safety Stock      (if ≤ 0 → no action)

Order Qty         = roundup_to_multiple( max(Net Req, EOQ where applicable), Order_Multiple )
                    subject to  ≥ MOQ,  ≤ Max_allowed (shelf-life cap),  ≤ vendor capacity

Release Date      = Required Date − Lead Time − Inbound QC days − Safety time
```

7.2 Rules that must be coded

- Low-level coding: process level 0, then level 1, and so on, so a component used at two levels is netted once
- Wastage/yield % applied at each BOM line
- Firm POs inside the frozen window are never auto-modified — only flagged with expedite / de-expedite / short-close recommendations
- **Pegging is mandatory.** Every suggestion traces to the demand line that caused it — click a suggested RM PO, see the FG forecast or channel PO behind it. Planners will not trust a black box, and untrusted suggestions get overridden into uselessness.
- Container/truck fill optimisation for long-haul and imported RM, with fill % shown
- Price-break optimisation: if a larger order drops the slab price and total landed cost falls, suggest it with the saving quantified — but never breach the shelf-life cap

7.3 Vendor allocation

Where multiple approved vendors exist: split by rate-contract allocation %, or optimise on a weighted score — landed price 40%, quality 25%, OTIF 25%, responsiveness 10% (weights configurable). **Never allocate to a vendor who is Blocked, has an expired certificate, has an open critical NCR, or is over agreed capacity.** Single-source items go on a risk register with a mandatory second-source development action.

7.4 PO workflow

```
MRP suggestion → Planner review (accept / modify / reject, reason mandatory)
→ PO draft → Approval matrix → Released to vendor
→ Vendor acknowledgement → ASN → GRN → QC → Putaway → 3-way match → Payment
```

Approval matrix (default, configurable):

Condition	Approver
PO ≤ ₹1,00,000	SCM Head
₹1,00,001 – ₹10,00,000	SCM Head + Finance
> ₹10,00,000	AP
Non-AVL vendor	AP, with QA sign-off

Condition	Approver
Qty > 25% above system suggestion	AP
Emergency / expedite PO	AP, with monthly post-facto review

- **True auto-release:** for C-class, rate-contracted, stable items below a value threshold, the PO releases to the vendor automatically with a daily digest to SCM Head. Enable only after 3 months of stable running — this is what "auto PO" should eventually mean.
- PO amendments versioned with vendor re-acknowledgement; open-PO ageing report; auto-reminders at LT−7, LT−3, LT and overdue; short-close and cancellation with reason codes

7.5 Capacity check (make items)

Before confirming a planned manufacturing order, check work-centre capacity by shift. If infeasible, flag and offer the options — overtime, alternate line, other plant, advance the build, de-prioritise a lower-margin SKU. Show the trade-off; the planner decides.

8. PROCUREMENT & VENDOR MANAGEMENT — M7

8.1 Sourcing

- **RFQ** to AVL vendors; quote comparison on **landed cost** (basic + freight + duty + GST credit adjustment + payment-term cost), award with a justification note
- **Rate contracts** with validity and escalation linked to a raw-material index where applicable (pulp, resin, SAP, fragrance oils, PE film) — index-linked contracts end the quarterly renegotiation drama
- **Should-cost models** for key RM/PM so negotiation is evidence-based
- New vendor onboarding: RFI → sample → sample QC approval → plant audit (checklist + score) → documents → trial order → performance review → AVL

8.2 Vendor portal

Vendors log in to: view and acknowledge POs; upload ASN with batch, MFG date, qty, vehicle and LR; upload COA and test certificates; **upload invoices** (this is the primary feed for M3); view GRN and rejection status; view payment status; view their own scorecard; receive NCRs and submit CAPA responses.

This single feature removes roughly 60% of SCM's follow-up phone calls and is the cleanest possible source for daily invoice capture. Build it in Phase 3, not "later".

8.3 3-way match and payment

`PO ⇔ GRN (accepted qty after QC) ⇔ Vendor invoice`. Default tolerance 0% on qty and rate; mismatches auto-block and queue. Auto-generate debit notes for rejected qty, short supply, rate difference and quality recovery. MSME 45-day clock tracked and alerted.

8.4 Vendor scorecard (auto, monthly)

Dimension	Weight	Measures
Quality	35%	Lot acceptance %, rejection PPM, NCRs, CAPA closure timeliness

Dimension	Weight	Measures
Delivery	30%	OTIF (on-time = within ±2 days of PO date; in-full ≥ 98% qty)
Cost	20%	Price variance vs rate contract, savings contribution
Responsiveness	10%	PO ack TAT, query TAT, ASN accuracy
Compliance	5%	Certificate validity, document completeness, audit score

Bands: A ≥ 85 (grow share) | B 70–84 (develop) | C 55–69 (watch + improvement plan) | D < 55 (exit plan). Auto-generate a quarterly business review pack per A/B vendor.

9. INBOUND — GATE TO PUTAWAY — M8

9.1 Flow (scanner-driven, this is a physical event — not an upload)

```
Gate entry (vehicle, LR, PO ref, time-in)
→ Weighbridge gross [bulk]
→ Unload, count vs ASN
→ GRN (scan PO → auto-fill; capture batch, vendor batch, MFG date, EXP date)
→ Stock posts to QUARANTINE — system-blocked from issue and sale
→ QC inspection auto-triggered
→ Disposition (Section 10)
→ Putaway to system-suggested bin, FEFO-aware
→ Gate time-out
```

9.2 Rules

- **No GRN without a PO.** Free samples, returns and job-work use dedicated document types — never a blank GRN.
- Over-receipt tolerance per item (default 0%, up to 2% for weight-based RM); beyond that needs approval
- **Every batch-controlled receipt captures batch + MFG + EXP. No exceptions.** If the vendor did not print it, it does not come in — this is the foundation of the entire QC module.
- Barcode/QR label printed at GRN for every pallet and case: item code, batch, qty, GRN ref, and a QC status colour (Red quarantine, Green released, Black rejected)
- Damage photos mandatory at unloading for claims; vehicle detention tracked (time-in to time-out) feeding vendor and transporter scorecards

9.3 Job work / third-party manufacturing

Challan-based issue of RM/PM to the converter, visibility of material at the vendor location, FG receipt against the challan, yield reconciliation, and the GST 143 one-year return clock. **QC on a job-work receipt is identical to normal incoming QC.**

10. QUALITY MODULE — M9

> Design principle: **quality status is a property of stock, not a piece of paper.** Nothing can be issued, dispatched or sold unless the ledger says the batch is RELEASED. QA can block; nobody else can unblock.

10.1 Stock quality states

State	Issue?	Sell?	Set by
QUARANTINE	No	No	System, on receipt or production
UNDER TEST	No	No	QC on sampling
RELEASED	Yes	Yes	QA approver only
CONDITIONAL RELEASE	Restricted	Restricted	QA Head + AP
ON HOLD / MRB	No	No	QA, Production, or a complaint
REJECTED	No	No	QA
RETURN TO VENDOR	No	No	QA + SCM
SCRAP	No	No	QA + Finance
RETENTION	No	No	QA — locked, non-transactable

Every transition logs user, timestamp, reason and e-signature. **Backdating is impossible; corrections are reversal entries, never edits.**

10.2 Quality masters

Parameter master — name, type (Variable / Attribute / Visual), method/SOP reference, instrument required, unit, specification (min / max / target / nominal ± tolerance), acceptance criteria, criticality (**Critical / Major / Minor**), CTQ flag, in-house vs external lab, test TAT.

Inspection plan — item + stage (IQC / IPQC / FQC / PDI) → parameters + sampling plan + frequency + inspector qualification + record template. Versioned, effective-dated, QA-approved. **No item goes Active without one.**

Sampling plan master

- Attribute sampling per **IS 2500 Part 1 / ISO 2859-1 (ANSI Z1.4)**, General Inspection Level II default; AQL by criticality — Critical 0.065, Major 1.0, Minor 2.5 (configurable)
- **The system computes sample size and Ac/Re from lot size and AQL. The inspector cannot choose it.**
- Automated normal / tightened / reduced switching: 10 consecutive accepted lots → reduced; 2 rejected out of 5 → tightened; 5 consecutive rejected on tightened → **suspend that vendor for that item** and escalate
- Variable sampling for weight and dimensional parameters; skip-lot for A-grade vendors after 20+ clean lots

Instrument master — ID, make/model, range, least count, location, calibration frequency, last and next calibration date, agency, certificate, traceability to national standards. **An instrument past its calibration date is auto-blocked — results cannot be entered against it.** Alerts at 30/15/7 days.

Personnel qualification — inspector-wise qualified parameters, training records, re-qualification dates. Only qualified inspectors can record results for a parameter.

Reference samples — golden samples, colour standards, fragrance references (critical for ALMA), approved artwork masters, with validity and controlled issue.

10.3 Incoming QC (IQC)

Auto-triggered on GRN:

20. System creates an inspection lot (one batch, one item, one GRN)
21. System computes sample size and generates a random sample-draw instruction (which cartons)
22. Inspector draws samples on the mobile app, scanning pallet labels
23. Results entered **parameter-wise with actual values**, not just pass/fail — you need the data for SPC and vendor trending. Photos for visual defects.
24. Vendor COA compared to in-house results; deviation beyond tolerance auto-flags. **A vendor COA is never a substitute for testing on Critical parameters.**
25. External lab: test request raised, TAT tracked, report uploaded and linked
26. Auto-verdict against Ac/Re; QA approver e-signs the disposition

Category parameter sets to configure — QA Head to finalise specs and confirm current BIS/standard revisions:

- **Baby diapers / pants (Babio):** layer-wise GSM, absorbency and acquisition time, rewet, retention capacity, leg-cuff elasticity, tape and side-seam peel strength, dimensions, SAP content, surface pH, dermatological safety certificate, microbial limits, pack integrity, artwork and MRP declaration
- **Wet wipes (Novel / Medibath):** GSM, sheet size, lotion pick-up %, pH, microbial limits (TAMC/TYMC; absence of *P. aeruginosa*, *S. aureus*, *E. coli*, *C. albicans*), preservative efficacy challenge test on any formulation change, lid and flap seal integrity, dry-out test, alcohol content where applicable
- **Baby shampoo / lotion / head-to-toe wash (new SKUs):** pH, viscosity, specific gravity, appearance, colour, odour vs reference, active matter, foam height and stability, microbial limits, preservative assay, stability (accelerated 40°C/75% RH for 3 months plus real-time), pack compatibility, tear-free claim substantiation on file
- **Perfumes (ALMA):** fragrance vs reference by blind panel, alcohol content, specific gravity, refractive index, colour, clarity, fill volume, spray rate, valve and actuator function, leak test, batch-to-batch odour consistency
- **Incontinence (Medibath):** absorbency, rewet, barrier integrity, odour control, adhesive performance, dimensions, microbial limits
- **Packaging material:** dimensions, GSM/thickness, **print vs approved artwork version** (a wrong MRP, net quantity or claim on pack is Legal Metrology and consumer-law exposure), colour ΔE, bursting strength, seal strength, barcode scannability grade, shipper drop test
- **All FG:** Legal Metrology declarations (MRP, net qty, mfg date, expiry, manufacturer address, customer care, country of origin), applicable BIS/CDSCO marking, GTIN verification against item master

10.4 In-process QC (IPQC)

- **Line clearance before every batch:** digital checklist — previous batch cleared, line cleaned, correct laminate/artwork loaded, correct coding set, machine parameters set, weights verified. **The MO cannot be started in the system until line clearance is signed.**
- Defined-frequency checks: fill weight/volume, seal integrity, coding legibility, dimensions, appearance, machine parameters
- Automatic control charts ($\bar{X}$-R for variables, p/np for attributes) with **Western Electric run rules** — alert supervisor and QA *before* a spec limit is breached
- Interlock: two consecutive out-of-spec checks → auto line-hold flag requiring supervisor acknowledgement
- **Bulk/compounding release** for lotions, shampoos and perfumes — pH, viscosity, odour, actives. Bulk cannot be filled until released.
- In-process rejection captured by defect code and machine, feeding Pareto and OEE quality rate

10.5 Final QC and Pre-Dispatch Inspection

- FQC on the packed batch before release
- **Channel-specific PDI:** carton labelling, FNSKU/ASIN label correctness, GS1 barcode scan test, shipper markings, palletisation, residual shelf life vs that channel's minimum, units per case, ASN match, MRP sticker compliance

- **Dispatch is hard-blocked if:** batch is not RELEASED, residual shelf life is below the channel minimum, or the PDI checklist is incomplete. A block, not a warning.

10.6 Batch record and release

- Electronic BMR and BPR auto-assembled from the transactions: RM/PM batches issued, quantities, operators, machine, process parameters, in-process results, yield, deviations, packaging reconciliation
- **Reconciliation:** theoretical vs actual yield against limits; outside limits requires a deviation and investigation before release
- **Release checklist:** BMR complete, all tests passed, deviations closed, packaging reconciled, artwork verified, retention sample kept, COA generated → QA Head e-signature → stock moves QUARANTINE → RELEASED
- Auto-generated **Certificate of Analysis** per batch, PDF, shareable with institutional buyers and marketplaces

10.7 NCR, deviation, MRB, CAPA

NCR raised from incoming, in-process, final, complaint or audit: source, item, batch, qty affected, description and photos, criticality, responsibility (vendor / production / warehouse / design), immediate containment.

MRB — quorum QA + Production + SCM, plus AP above a value threshold. Dispositions: `Accept as-is` (justified) | `Accept with deviation` (documented, time-bound, **never for Critical parameters**) | `Rework` (QA-approved procedure, 100% re-inspection) | `Downgrade` (B-grade / institutional, with brand approval) | `Return to vendor` (auto debit note) | `Scrap` (destruction record with photo evidence).

Deviation management — planned and unplanned, risk assessment, QA approval before proceeding, closure with effectiveness check.

CAPA — mandatory for every Critical NCR and every repeat Major NCR. Structure: problem → containment → **root cause via recorded 5-Why or Fishbone**, not a free-text box → corrective action → preventive action → owner → due date → **verification of effectiveness with evidence** → closure. Overdue CAPA dashboard escalates to AP weekly. A repeat NCR on the same cause within 6 months reopens the CAPA and marks effectiveness failed.

8D report generator for customer and marketplace-facing quality issues.

10.8 Complaints, market feedback, recall

- Ingest from Amazon/Flipkart/Meesho reviews and return reason codes, quick-commerce complaints, D2C support, customer care, distributors
- Log with batch — put a QR on pack so consumers can report against a batch — plus defect category and severity
- Auto-link complaint → batch → BMR → RM lots → vendor, and **show every other batch made from the same RM lot in one click**
- Trend complaints per million units by SKU, batch, line, shift and vendor lot; track consumer response TAT
- **Recall / market withdrawal module:** define scope by batch or RM lot, generate the full distribution list (channel, FC, distributor, quantity, dispatch date), issue notices, track recovered vs dispatched, produce a reconciliation report. **Run a mock recall every 6 months — the trace must complete in under 10 minutes and export cleanly for audit.**

10.9 Traceability

Backward: FG batch → BMR → all RM/PM batches consumed → GRN → vendor → vendor batch → COA → inspection results.

Forward: RM lot → all FG batches → all dispatches → channel / FC / distributor → invoice → (D2C) customer order.

One screen, both directions, exportable. This is the feature that protects the company if a baby-care batch is ever questioned.

10.10 Audit and compliance support

Document control (SOPs and formats — versioned, effective-dated, read-acknowledgement tracked per employee); internal audit scheduler with checklists, findings and CAPA linkage; an **external audit pack generator** (customer, marketplace, BIS/CDSCO, ISO 22716 cosmetic GMP) that pulls SOPs, calibration, training, batch records and complaint trends into one export. Quality records are append-only; retention shelf life + 5 years for baby care.

11. MANUFACTURING — M10

- **MPS** from the constrained consensus forecast plus firm orders → weekly plan by line and plant
- Manufacturing Order → shift job cards → material issue (**scanned physical issue for A-class RM**; backflush only for low-value consumables) → production confirmation → FG receipt into QUARANTINE
- Material reconciliation per MO: issued vs consumed vs returned vs scrapped; yield vs standard, with investigation above a variance threshold
- **OEE by machine and line** = Availability × Performance × Quality, with downtime reason codes (breakdown, changeover, material shortage, manpower, quality hold, no plan) and a weekly downtime Pareto
- Changeover tracking and SMED improvement log
- Preventive maintenance schedule, breakdown log, spares consumption linked to spares inventory and auto-reorder
- Shift productivity and output per manhour
- Scrap and wastage captured by reason at every stage, valued and trended — this is direct margin

12. INVENTORY MANAGEMENT & CONTROL — M8

12.1 Core

- Real-time stock by item × batch × location × bin × quality state, always derived from the ledger
- **No manual quantity edits anywhere.** Every change is a posted document with a reason code.
- **FEFO enforced** for shelf-life items (FIFO otherwise). The system nominates the batch at picking; override needs supervisor approval and a reason.
- Reservation against confirmed channel POs so the same stock is never promised twice
- Goods-in-transit visibility: plant → FC, plant → distributor, vendor → plant, plant → job worker

12.2 Counting

- **Cycle counting by ABC** — A monthly, B quarterly, C half-yearly. System generates the list; **blind count** (expected qty hidden on the device); variance report; recount above tolerance; adjustment approval matrix (> ₹25,000 → AP)
- Full physical stock take with freeze, count, reconcile, post
- **Inventory accuracy % published monthly by location and by counter.** Target ≥ 99% for A-class.

12.3 Expiry and ageing

- Ageing buckets (0–30 / 31–60 / 61–90 / 91–180 / 180+ days), value-weighted

- Expiry workflow: 6 months residual → flag to Sales; 4 months → discount/combo approval; 3 months → institutional or liquidation channel; 2 months → write-off provision; expired → block and scrap with destruction certificate
- Provisioning report for Finance on slow, near-expiry and dead stock

12.4 Returns

- Marketplace returns (customer, unfulfillable, FC-damaged), distributor returns, D2C returns
- Receipt → **mandatory QC grading**: A (resaleable / repack), B (secondary / institutional), C (scrap)
- Repack work order for A-grade, retaining traceability to the original batch
- Return-reason analytics by SKU and channel, feeding product and packaging CAPA
- **Reconciliation of marketplace return claims vs physically received** — marketplaces under-return, and the gap is claimable money

13. OUTBOUND & CHANNEL FULFILMENT — M11

13.1 Channel PO ingestion

The existing SCM PO tracker is absorbed into this module. Do not run two systems.

Channel	Ingestion	Notes
Amazon	SP-API (Vendor PO / FBA inbound)	FNSKU labelling, shipment plan, IPI and restock limits
Flipkart	Seller/Vendor API	FC-wise allocation, appointments
Meesho	API / portal export	High volume, price-sensitive, returns-heavy
Blinkit / Zepto / Instamart / BigBasket	Portal export or API; RPA fallback	Short PO windows, appointment slots, strict fill-rate penalties
Tata 1mg, FirstCry	Portal export	Batch and expiry mandatory
Shopify D2C	Admin API	Order-level, real-time
GT distributors	Distributor app / upload	Primary, secondary and closing stock

Per PO capture: PO number, SKU, qty, rate, delivery window, destination FC, appointment status, ASN status, dispatched, received, short/excess, **fill rate**, penalty exposure, and a reason code for every shortfall.

13.2 Fulfilment flow

```
Channel PO → ATP check (RELEASED stock only, FEFO, channel shelf-life rule)
→ Allocate & reserve → Book appointment → Pick list (wave/batch, scan-verified)
→ Pack (case labels, channel labels, shipper marking) → PDI checklist
→ Tax invoice + e-invoice IRN + e-way bill → ASN transmitted
→ Dispatch (vehicle, LR, transporter) → POD → Received-qty reconciliation
```

13.3 Controls

- **Cannot dispatch:** unreleased batch, short-dated stock, un-appointed shipment, ASN/physical mismatch, incomplete PDI

- Daily fill-rate dashboard by channel with root cause on every miss — no stock / short-dated / appointment miss / QC hold / logistics
- Penalty and chargeback tracker per channel with dispute status
- Transporter scorecard: on-time delivery, damage rate, cost per case, detention

13.4 Quick commerce replenishment — priority, this is where the growth is

- Dark-store / city-warehouse sell-out ingestion
- Auto-replenishment at 3–7 days cover, 2–3 dispatches weekly, tuned per city
- New-city launch pipeline: seed stock calculation, ramp curve, review at weeks 2 / 4 / 8
- Alerts on: platform-level OOS, listing deactivation, PO not received when expected, sudden sell-out spike (a competitor OOS opportunity)

13.5 GT / distributor — UP rollout onward

Primary sales, secondary sales, distributor closing stock, days of inventory at distributor, scheme and claim management, and a stock-and-sales report feeding the forecast. **Never forecast GT on primary sales — that is a pipeline number, not demand.**

14. COSTING — M12

- **Landed cost** per receipt: basic + freight + insurance + duty + clearing + other, apportioned by value, weight or volume, posted to the valuation ledger
- Variance analysis: purchase price variance, usage variance, yield variance, overhead absorption
- **Gross margin per SKU per channel:** net realisation − landed COGS − channel commission − fulfilment − ads − returns cost. Several "growth" SKUs are usually loss-making and nobody knows it. Put this in front of AP monthly.
- Inventory valuation by item, batch, node, quality state and ageing; provision for slow, non-moving and expired stock
- Payables ageing, MSME compliance, quality debit notes auto-posted
- GST outputs per Section 4.5

15. CONTROL TOWER — M13

15.1 Role dashboards

AP / Founder — one screen plus a 07:00 IST WhatsApp/email digest: yesterday's sales by channel vs plan | inventory value and cover | SKUs out of stock (A-class in red) | channel fill rate | open critical quality issues | POs awaiting my approval | cash committed in open POs | **ingestion exception queue size** | top 5 exceptions needing a decision

SCM Head — MRP suggestions pending, PO status, vendor OTIF, inbound pipeline, stock-out risk list, ageing and dead stock, redistribution suggestions, unmatched invoices

QA Head — lots pending inspection, batches pending release, open NCR/deviation/CAPA with ageing, calibration due, complaint trend, vendor PPM, audit findings

Plant Head — plan vs actual, OEE, downtime Pareto, yield and wastage, manpower productivity, material shortage list

KAM — PO fill rate, appointment status, stock at their FCs, near-expiry in their channel, penalty exposure

15.2 Alert engine

Configurable rule → channel (in-app / email / WhatsApp) → recipient → escalation ladder L1 owner → L2 head → L3 AP on SLA breach. **Every alert must have an owner and an action.** Cap alerts per person per day and batch low-priority ones into a digest, or they become noise and get ignored.

15.3 Reporting

Standard report library (60+), ad-hoc query builder, scheduled subscriptions, Excel/PDF export on everything, and drill-down from every report to the source document.

16. NON-FUNCTIONAL — M14

16.1 Roles and segregation of duties

Roles: Admin, SCM Head, Planner, Buyer, Store Keeper, Warehouse Operator, Production Supervisor, QC Inspector, QA Approver, QA Head, Plant Head, KAM, Finance, Auditor (read-only), Vendor (portal), Distributor (portal), Founder (all-read + approvals).

Enforced separations: PO creator ≠ PO approver | QC inspector ≠ batch releaser | stock counter ≠ adjustment approver | vendor bank-detail change requires dual approval.

16.2 Data integrity

Global append-only audit trail; no hard deletes; e-signature on quality dispositions and approvals; **server-side timestamps only, never the client clock**.

16.3 Performance

Page load < 2s at p95 | scan transactions < 500ms | full MRP run < 15 min | 200 concurrent users | 5 years of history without degradation | 99.5% uptime in business hours.

16.4 Mobile

PWA, offline-capable with a sync queue, camera and handheld barcode scanning, covering: gate entry, GRN, putaway, picking, cycle count, stock transfer, QC sample draw and result entry, line clearance, production confirmation.

16.5 Security, backup, DR

Daily automated backup with point-in-time recovery and a **quarterly tested restore — an untested backup is not a backup**. Encryption in transit and at rest. MFA for admin and approver roles. IP allowlist for portal admin. Documented DR plan, RTO 4h / RPO 1h.

16.6 Integrations

Build an integration layer, not point-to-point connections. Every integration gets retry with exponential backoff, a dead-letter queue, failure alerting, and a manual reprocess screen. Covers: Amazon SP-API, Flipkart, Meesho, quick-commerce platforms, Shopify, GSP (e-invoice + e-way bill), Tally export, courier/transporter APIs, WhatsApp Business API, external lab uploads.

17. CORE DATA MODEL (indicative)

```
-- Platform
company, user, role, permission, audit_log, workflow_def, workflow_instance,
number_series, attachment, notification, integration_log

-- Masters
item_master, item_channel_map, item_uom_conv, bom_header, bom_line, routing, ecn
vendor_master, vendor_item, rate_contract, vendor_certificate, vendor_score
customer_channel, warehouse, zone, bin, tax_master, mrp_history

-- Transactional core
document_header, document_line, stock_ledger_entry, valuation_ledger_entry,
stock_balance_mv, batch_master, period_lock, ewaybill, einvoice_irn

-- Ingestion
uploaded_document, extraction_result, item_mapping_learned, match_result,
exception_queue, gstr2b_line, recon_result

-- Planning
forecast_version, forecast_line, demand_event_calendar, forecast_accuracy,
inv_policy(item,node,abc,xyz,fsn,service_level,ss,rop,eoq,max,doc_target),
mrp_run, mrp_suggestion(pegging_ref), capacity_plan

-- Procure to pay
rfq, quote, po_header, po_line, po_amendment, asn, gate_entry,
grn_header, grn_line, purchase_invoice, debit_note, three_way_match

-- Quality
qc_parameter, qc_inspection_plan, qc_plan_line, qc_sampling_plan,
qc_instrument, qc_calibration, qc_personnel_qualification, reference_sample,
qc_inspection_lot, qc_result, qc_disposition, ncr, ncr_action, capa,
deviation, mrb_decision, batch_record, batch_release, coa, retention_sample,
complaint, recall_event, recall_line, sop_document, audit_finding

-- Make
mps, mo_header, mo_line, job_card, material_issue, production_receipt,
line_clearance, downtime_log, scrap_log, oee_daily, maintenance_schedule

-- Deliver
channel_po, atp_check, allocation, pick_list, pack_list, pdi_checklist,
dispatch, tax_invoice, credit_note, fill_rate_log, penalty_log,
return_receipt, return_grading, stock_transfer, cycle_count, count_variance,
stock_adjustment, distributor_secondary_sales
```

18. PHASED DELIVERY

Longer than v1.0 because the transactional core is now ours to build. **Do not compress Phase 1 — everything else stands on it.**

Phase	Weeks	Scope	Definition of done
0 — Master data & setup	1–5	Master data cleansing and migration, item/BOM/vendor/warehouse/bin setup, environments, repo, baseline metrics	Masters signed off by SCM + QA Heads; 100% of active SKUs have a BOM and an inspection plan
1 — Transactional core	4–14	Document engine, numbering, **stock ledger**, valuation, period lock, workflow engine, audit framework, roles, GST documents, e-invoice, e-way bill, Tally export	Ledger reconciles to zero across all locations under a load test; a full month closes cleanly

Phase	Weeks	Scope	Definition of done
2 — Inbound, inventory, mobile	12–20	Gate, GRN, batch capture, putaway, bins, FEFO, STO, cycle count, scanner PWA	Inventory accuracy ≥ 97% at Mysuru; zero manual quantity edits possible
3 — Quality module	14–24	Full Section 10 — plans, sampling, IQC/IPQC/FQC, NCR/CAPA, batch release, COA, traceability, recall	Mock recall completes in < 10 min; no unreleased batch can be dispatched
4 — Ingestion & procure-to-pay	18–26	Invoice upload pipeline, OCR, matching, exception queue, GSTR-2B recon, RFQ, PO lifecycle, 3-way match, **vendor portal**	90% of vendor invoices auto-matched; top 20 vendors live on the portal
5 — Planning & auto-PO	24–32	Forecasting, consensus, policy engine, MRP, PO suggestions, approvals	3 consecutive MRP runs accepted with < 20% manual override
6 — Outbound & channels	28–36	Channel PO ingestion, ATP, allocation, pick-pack-dispatch, ASN, PDI, fill rate, QC replenishment	Daily fill rate live for all channels; PDI blocking enforced
7 — Manufacturing, costing, control tower	32–42	MO, BMR/BPR, OEE, maintenance, landed cost, SKU × channel margin, dashboards, alerts	AP daily digest running; SKU-channel margin published monthly
8 — Manesar, GT, hardening	38–48	Second plant, distributor module, performance tuning, DR test, documentation	Both plants live; DR restore tested successfully

Parallelisation: Phases 2 and 3 overlap from week 14. Phase 4 ingestion can start at week 18 in parallel with QC. The QA Head's specification work (Section 10.3) must start in week 1 — it is the longest-lead input in the project and it is not a developer task.

Interim reality: between now and Phase 5 (roughly month 6), planning continues on the current spreadsheets. Do not switch off what works until the replacement is proven in parallel for one full month.

19. ACCEPTANCE CRITERIA — no go-live without these

27. The stock ledger sums to zero across all locations, and the derived balance table matches the ledger exactly after a full rebuild.
28. A batch that fails QC **cannot** be issued, dispatched, invoiced or sold — verified by attempting each in UAT.
29. Any FG batch traces backward to every RM/PM lot and vendor, and forward to every dispatch and channel, in **under 10 minutes**, exportable.
30. MRP produces a suggested PO that a planner can peg back to the exact demand line that caused it.
31. Safety stock, ROP and EOQ recalculate using **measured** lead times from GRN-vs-PO dates, not master-data assumptions.
32. Inspection sample size is computed by the system from the AQL plan; the inspector cannot alter it.
33. An instrument past its calibration date blocks result entry.
34. Segregation of duties holds — no single account can create and approve a PO, or inspect and release a batch.
35. A month closes and locks; a back-dated posting into it is rejected.
36. Uploaded invoices auto-match to PO and GRN at ≥ 90% for the top 20 vendors, with the remainder correctly queued as exceptions.
37. Mock recall executed end-to-end with a reconciliation report.
38. Backup restored successfully into a clean environment.

20. RISKS

Risk	Impact	Mitigation
Stock ledger designed as a mutable balance field	Unrecoverable data corruption by year two	Ledger-first design, code review gate in Phase 1, load test in acceptance
Master data garbage-in	Whole system distrusted, project dies	Phase 0 gate, named owner per master, no Phase 1 start without sign-off
Invoice-upload treated as a substitute for scanning	No batch, no traceability, no QC, no MRP	Section 3.4 is binding; physical events are scanned, documents are uploaded
Team bypasses the system (WhatsApp POs, paper QC)	Parallel truth	Approvals only in system; no payment without system 3-way match; weekly bypass report to AP
Building a general ledger	6 months lost to accounting	Explicitly out of scope; export to Tally
Over-engineering v1	Never ships	Hard phase gates; ship Phases 1–3 even if later phases slip
QC seen as blocking production	Pressure to override	Only QA Head can release; every override logged and reviewed by AP monthly
Single developer dependency	Continuity risk on a 48-week build	Git from day 1, mandatory module documentation, minimum 2 people across the codebase, no undocumented work
Integration breakage at channels	Fill-rate misses	Retry, DLQ, alerting, and a manual upload fallback for every channel
Hosting on the current server	Data loss	Migrate to managed cloud with tested PITR before Phase 2

21. WHAT I NEED FROM THE TEAM — next 7 days

39. **Sumukha:** confirm stack, environments, repo, team composition, and give a phase-wise estimate against Section 18. Specifically: your design for the stock ledger (Section 4.2) before any other code is written, and an honest read on whether the current team can carry a 48-week build or we need to add people.
40. **SCM Head:** name a data owner per master; deliver the current item, vendor and BOM extracts for cleansing.
41. **QA Head:** finalise parameters, specification limits and sampling AQLs per category in Section 10.3. **Start today** — this is the longest-lead input in the project.
42. **Plant Heads:** bin and zone layout for Mysuru and Manesar, including physically demarcated quarantine and rejection areas.
43. **Finance:** confirm the approval matrix values, the inventory holding-cost %, and the Tally export format.
44. **All function heads:** baseline the metrics in Section 0.1. We cannot claim improvement without a starting number.

Novel Group — v2.0, BOS-native. This is the build baseline. Changes to be raised as versioned change requests against this document.

— AP